Zikai Xiong

+86-189-1657-7653 • zkxiong16@fudan.edu.cn • ZikaiXiong.github.io

EDUCATION BACKGROUND

Fudan University **Shanghai, China**
B.S. in Mathematics and Applied Mathematics *Sept. 2016 - Jun. 2020*
GPA: 3.84/4.00 GRE Mathematics Subject Test: 970 (99%)

Related Course
Statistics: probability theory, stochastic process, simulation, statistical learning;
Optimization: convex optimization, linear programming, non-linear programming, numerical analysis;
Mathematics: functional analysis, complex analysis, real analysis, topology, algebra, differential geometry;

Tools and Technologies
Proficient Languages: Matlab, C++, R; Optimization Software: CPLEX, Gurobi, Mosek;

RESEARCH EXPERIENCE

Smart Cross-Over for Linear Programming
***Primary Contributor**, Advised by **Prof. Yinyu Ye** (Stanford)* *Jun. 2019 - Present*
- Utilized the spanning tree structure to develop several smart crossover methods for network flow problems, Wasserstein barycenter problems, and Markov Decision Problems;
- Proposed a new exact OT problem algorithm that is tens of times faster than the best existing exact algorithm and hundreds of times faster than the state-of-art commercial solver Gurobi;
- Will submit to a prestigious peer-reviewed journal.

Interior-point Methods Strike Back: Solving the Wasserstein Barycenter Problem (Accepted by NeurIPS 2019)
***Co-Corresponding Author**, Advised by **Prof. Yinyu Ye**(Stanford) and **Prof. Dongdong Ge**(SUFE)* *Dec. 2018 - Aug. 2019*
- Developed a matrix-based adaptive alternating interior-point method (MAAIPM) to solve the large-scale linear programming sub-problems of Wasserstein barycenter, which is an important and popular problem in both logistics and A.I. community;
- Reduced the complexity of each iteration from standard IPM's $O(N^3m^4)$ to MAAIPM's $O(Nm^3)$ and attained a balance between accuracy and speed;
- Link: https://papers.nips.cc/paper/8913-interior-point-methods-strike-back-solving-the-wasserstein-barycenter-problem.

Effect of Hidden-city Ticketing in Revenue Management
***Primary Contributor**, Advised by **Prof. Bo Jiang** (SUFE)* *Apr. 2018 - Present*
- Proposed a setting that allows consumers to buy products according to certain probability determined by choice models including the Multinomial Logit (MNL) Choice Model
- Proved the $1/2$ upper bound of customer hidden-city ticketing practice's effect on airlines' revenue via a 2-stage price-modification, although exactly solving the dynamic programming formulation is difficult;
- Some quantitative bounds can still be established even when only a part of passengers practice hidden-city ticketing;
- Target Journal: **Transportation Science** (The extended abstract will be updated in my personal website soon)

Dynamic Assortment Optimization with Changing Contextual Information
***Research Assistant**, Advised by **Prof. Xi Chen** (NYU) and **Prof. Yuan Zhou** (UIUC)* *Jun. 2018 - Oct. 2018*
- Completed the numerical studies and visualized the results:
  - Developed an approximation algorithm and an efficient greedy heuristic to solve the combinatorial optimization;
  - Implemented an upper confidence bound (UCB) based policy to simultaneously learn and make decision;
- Submitted to **Management Science**.

PROFESSIONAL EXPERIENCE

- Invited Talks
  - INFORMS Annual Meeting, *Interior-point Methods Strike Back: Solving the Wasserstein Barycenter Problem* *Oct. 2019*
  - Seminar of New Advances in Theory and Application of ADMM (hosted by SUFE), *Computing Wasserstein Barycenter Efficiently: A Structured Linear System and Customized Algorithms* *Mar. 2019*
- Teaching Assistants
  - *Stochastic Process and Financial Risk Analysis* (Prof. David Yao from Columbia University), graduate-level; and *Stochastic Modeling* (Prof. Peter Glynn from Stanford), graduate-level *Summer 2019*
    - Held office hour and graded assignments for over 100 students
  - *Linear and Non-linear Programming* (Prof. Dongdong Ge), undergraduate-level *Fall 2018*
    - Held weekly recitations for over 20 students

HONORS & AWARDS

- Fudan Outstanding Student Pacesetter Award *Nov. 2019*
  - The highest and most prestigious award for undergraduate students in Fudan University (10/13377 0.07%);
  - As the only recipient from the School of Mathematical Sciences in the past two years.
- National Scholarship, the highest annual scholarship for students ranking in the top 1% in the school *2018*
- National Scholarship, the highest annual scholarship for students ranking in the top 1% in the school *2017*
- First-Class Award of Huadong Cup National Mathematical Modeling Contest *Apr. 2018*
- First-Class Award of National College Students Mathematical Competition *Oct. 2017*
- National Basic Science Talent Training Plan *Sept. 2017 - Present*
  - For students who attend weekly seminars and give presentations every semester